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Studierichting: Informatica
Oefeningenreeks 7D nr. 1.3

Opgave: Bereken de Taylorpolynoom

$$f(x, y) = e^{x^2-y} \text{ in } (0, 0), \text{ graad } 3$$

$$f(0, 0) = e^0 = 1$$

Eerste orde partiële afgeleiden:

$$\frac{\partial f}{\partial x} = 2xe^{x^2-y} \Rightarrow \frac{\partial f}{\partial x}(0, 0) = 0$$

$$\frac{\partial f}{\partial y} = -e^{x^2-y} \Rightarrow \frac{\partial f}{\partial y}(0, 0) = -1$$

Tweede orde partiële afgeleiden:

$$\frac{\partial^2 f}{\partial x^2} = 2e^{x^2-y} + 4x^2e^{x^2-y} \Rightarrow \frac{\partial^2 f}{\partial x^2}(0, 0) = 2$$

$$\frac{\partial^2 f}{\partial y^2} = e^{x^2-y} \Rightarrow \frac{\partial^2 f}{\partial y^2}(0, 0) = 1$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = -2xe^{x^2-y} \Rightarrow \frac{\partial^2 f}{\partial x \partial y}(0, 0) = \frac{\partial^2 f}{\partial y \partial x}(0, 0) = 0$$

Derde orde partiële afgeleiden:

$$\frac{\partial^3 f}{\partial x^3} = 4xe^{x^2-y} + 8xe^{x^2-y} + 8x^3e^{x^2-y} = (12x + 8x^3)e^{x^2-y} \Rightarrow \frac{\partial^3 f}{\partial x^3}(0, 0) = 0$$

$$\frac{\partial^3 f}{\partial y^3} = -e^{x^2-y} \Rightarrow \frac{\partial^3 f}{\partial y^3}(0, 0) = -1$$

$$\frac{\partial^3 f}{\partial x^2 \partial y} = \frac{\partial^3 f}{\partial y \partial x^2} = \frac{\partial^3 f}{\partial x \partial y \partial x} = -2e^{x^2-y} - 4x^2e^{x^2-y}$$

$$\Rightarrow \frac{\partial^3 f}{\partial x^2 \partial y}(0, 0) = \frac{\partial^3 f}{\partial y \partial x^2}(0, 0) = \frac{\partial^3 f}{\partial x \partial y \partial x}(0, 0) = -2$$

$$\frac{\partial^3 f}{\partial y^2 \partial x} = \frac{\partial^3 f}{\partial x \partial y^2} = \frac{\partial^3 f}{\partial y \partial x \partial y} = 2xe^{x^2-y}$$

$$\Rightarrow \frac{\partial^3 f}{\partial y^2 \partial x}(0, 0) = \frac{\partial^3 f}{\partial x \partial y^2}(0, 0) = \frac{\partial^3 f}{\partial y \partial x \partial y}(0, 0) = 0$$

Taylorpolynoom:

$$T_3(x, y) = 1 - y + \frac{1}{2!}(2x^2 + y^2) + \frac{1}{3!}(-2x^2y - y^3)$$

$$T_3 = 1 - y + x^2 + \frac{y^2}{2} - \frac{x^2y}{3} - \frac{y^3}{6}$$

References